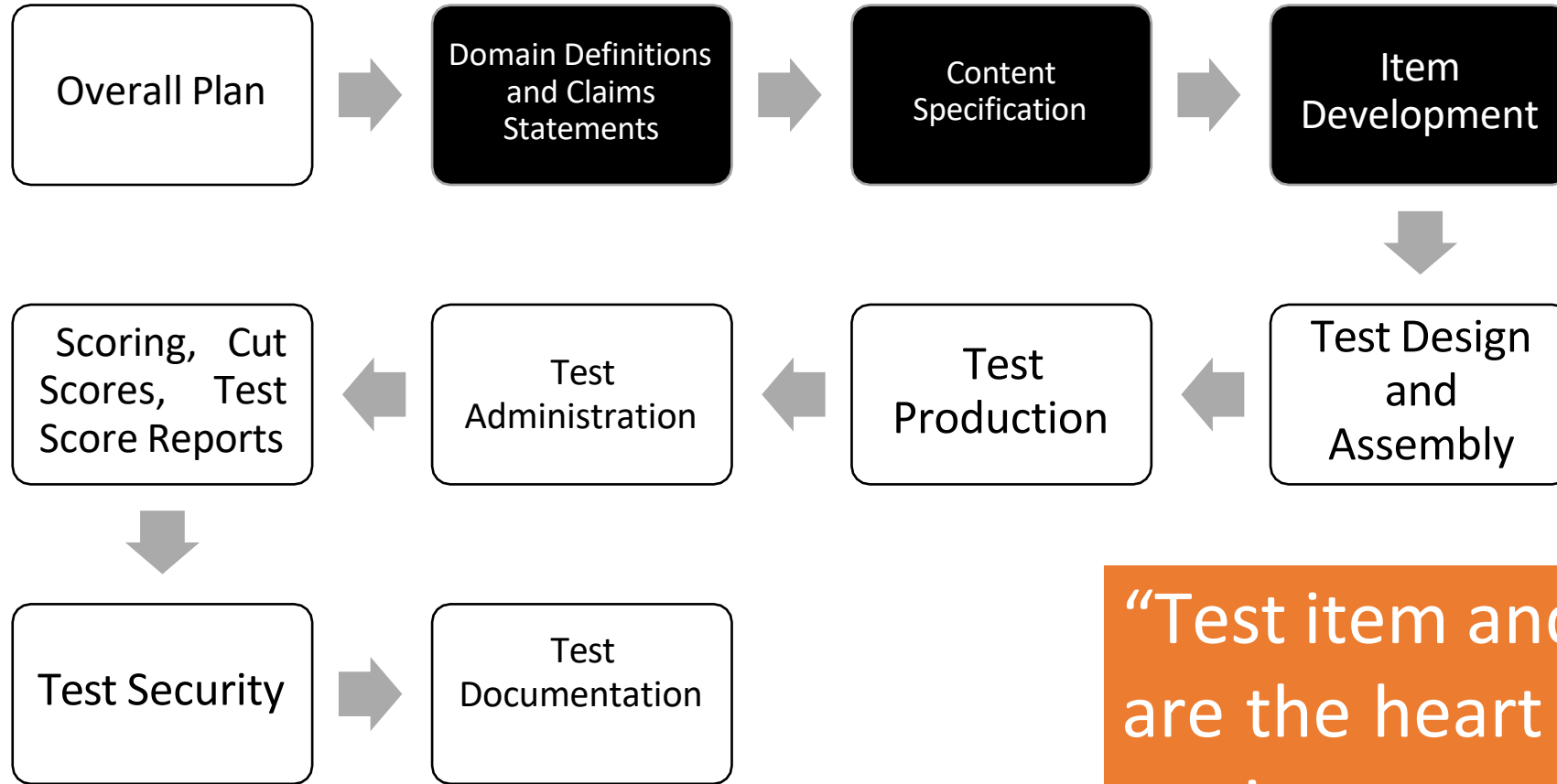


GUIDE TO WRITING COGNITIVE TEST ITEMS

Cognitive tests

- A test that measures a person's ability to perform a cognitive task.
- Cognitive tasks are made up of different groups of cognitive abilities that involve different cognitive processes.
- Cognitive processes are mental processes that produce certain responses in individuals, it can be related to the process of rerepresenting or changing external stimuli or stimuli images, knowledge, rules, etc. stored in short-term or long-term memory.

Test Development Stages



(Lane, Haladyna, Downing, 2016)

“Test item and tasks are the heart of any testing program”

Domain Definitions
and Claim
Statements



Describe the construct
domains to be tested and
define what will be measured
from the test-taker view.

Content specification

“documentation of the purpose and intended uses of the test, as well as detailed decisions about content, format, test length, psychometric characteristics of the items and test, delivery mode, administration, scoring, and score reporting” (The Standards, AERA et al., 2014)

Provide operational definitions of the test content and cognitive demands of the tasks assigned in the assessment

Used as a guide to compile sampling from assigned domains.

Domain definitions and test specifications provide content-based proof of validity



Construct domains to be tested (definition, definition, theory)



Expected cognition processes



Other things related to both, for example: difficulty level and number of items

*Test
Content
Specification*

Example of *Test Content Specification*

Broad Ability	Narrow Ability/Test specific	Type of content	Easy	Medium	Difficult
Fluid reasoning (Gf)	Induction-Verbal Analogy	Absence of X causes Y	2	3	2
		Part	2	3	2
		X Result on Y	3	3	2
		Levels	3	3	2
	JUMLAH	30	10	12	8
	Logical Reasoning	Barbara	3	2	-
		Celarent	2	4	-
		Darii	1	2	2
		Ferio	1	1	2

Item Development



Item Specification:

- The content domains and cognitive tasks provided include clues about construct descriptions and equivalent tasks on various items, clues about the level of cognitive complexity, the expected level of difficulty,
- Item format and item structure (item table/nomenclature)
- Item writing guide.
- A guide to writing style or format.
- Review process or criteria.
- Item selection criteria.

Contoh *Item Specification*

Subject Area:	Science
Cluster:	Laboratory and Field Work
Skill:	Recognize the role of observation and experimentation in the development of scientific theories.
Item Format:	Multiple Choice, four responses.
Item Type A:	Isolate and define a scientific problem or area for scientific study. 1) Stimulus Characteristics: a. The stimulus will direct the student to identify a statement that best defines a scientific problem or topic for investigation. b. The stimulus will provide a thorough description of observational data from which the student will discern the area to be investigated. c. The material should neither be so well known that the student is likely to be familiar with actual scientific work in the area nor so technical that specialized knowledge is required.

2) Response Options:

- a. The correct answer will be a sentence or brief paragraph that presents an accurate and clearly defined statement of the scientific problem or area of study.
- b. Distractors will be sentences or brief paragraphs that fail to define the scientific problem or area of study accurately, and which may be incorrectly or too broadly stated or may describe material that is irrelevant or outside the scope of the observation.

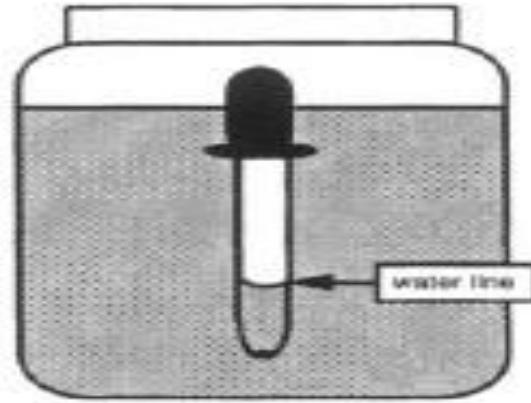
Sample Item: A researcher studying the use of radiation to slow the spoilage of vegetables harvested four dozen tomatoes of a variety that had been grown under identical conditions and were approximately the same size and weight. After randomly selecting them for placement in sterile racks containing a dozen each, she subjected two racks to a fixed amount of radiation. Then she placed one irradiated rack and one that had not been irradiated in refrigerators which maintained the same temperature. She stored the other two racks, one of which had been irradiated, at room temperature. She checked all four racks every six hours for signs of spoilage. When she found signs of spoilage in at least three of the tomatoes in a rack, she recorded the elapsed time and disposed of the tomatoes in that rack. The researcher found that the irradiated tomatoes spoiled at the same rate as those that were not irradiated. Repetition of the experiment gave the same results.

Which variable would it be best for the researcher to alter in her follow-up experiment?

- A. amount of radiation
- B. storage temperature
- C. number of tomatoes
- D. size of tomatoes

Item Nomenclature

Use the illustration and read the passage below to answer the question.



Charley did an experiment in which he floated an eyedropper in a jar of water. He left the jar on a shelf for one hour. Then he recorded the level of water in the eyedropper. Next, he heated the jar containing the eyedropper for 30 seconds.

Which variable did Charley change in this experiment?

- A. kind of eyedropper
- B. level of water in the jar
- C. amount of time between temperature measurements
- D. temperature of the air and water in the jar

Direction

Graphic

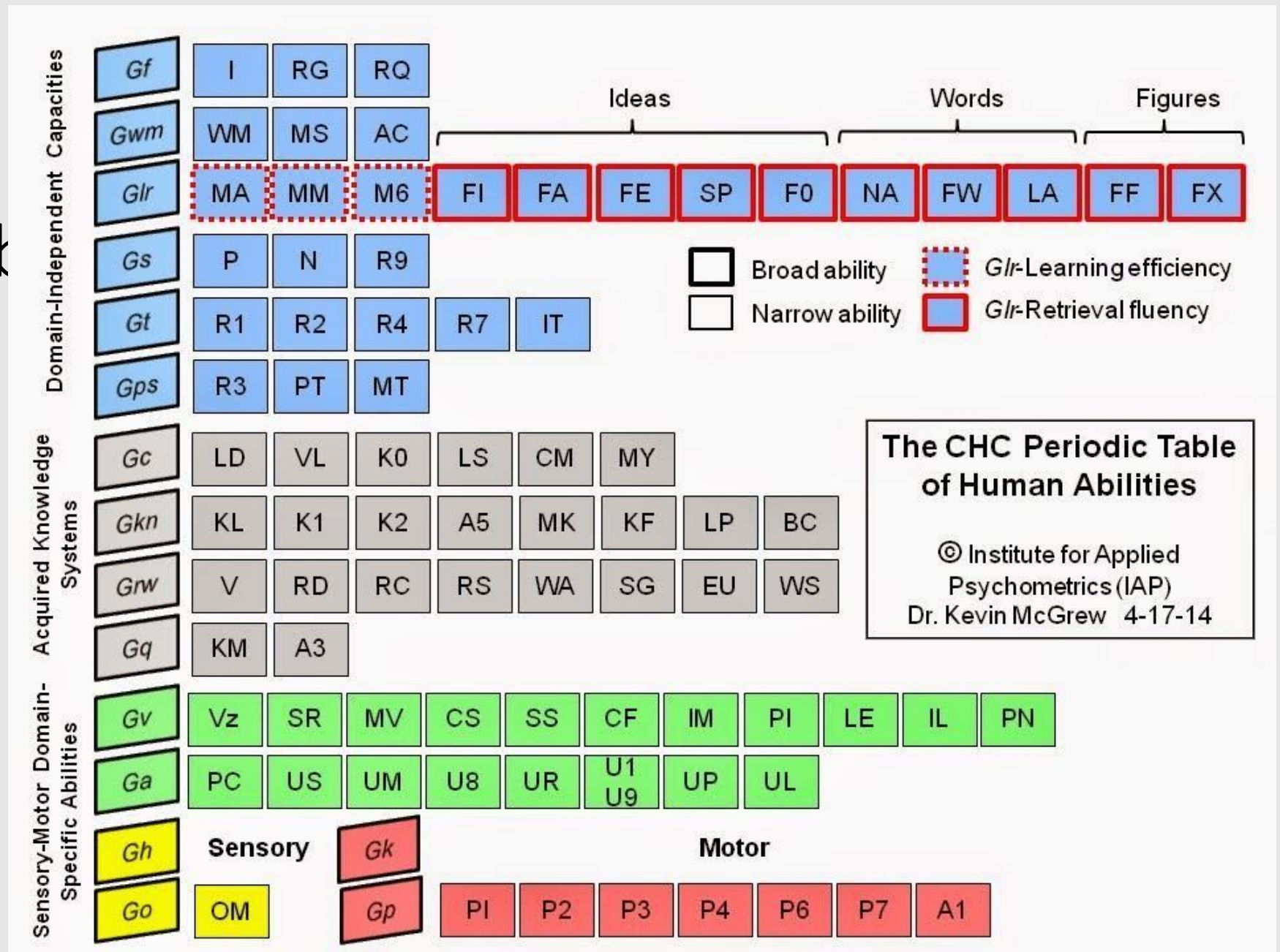
Text

Stem

Distractors

*Correct
Response*

CHC Taxonomy Periodic table



Item writing: content concerns

Limit each item to one content type and cognitive demands.

Use new material to measure higher thought processes.

Make sure the content of the items is independent of each other.

Test content that matters. Avoid content that is too specific and too general.

Avoid opinions.

Avoid tricky items.

Item
writing:
format
concerns

Format each
item
vertically
rather than
horizontally.

Item writing: style concerns

Edit and check items.

Make sure the language complexity is appropriate for the group to be tested.

Limit the reading length on each item to a minimum.
Avoid window dressing.

Item writing: writing the stems

Write down the core idea clearly and concisely in the stem section, not in the answer choice.

Use positive words on stems; not negative (e.g. NO, EXCEPT)

Item writing: writing the options (1)

Use only possible and differentiating answer options.

Make sure there is only one correct answer.

Put the correct answers variously across different items.

Sort answer choices in logical order or numerical.

Make sure each option is independent.

Avoid using options: everything is wrong, everything is right, and I don't know.

Use positive words in the answer choices; avoid negative words, such as, NO.

Item writing: writing the options (2)

Avoid giving clues to the correct answer:

- *Make sure the length of the answer choice is equivalent.*
- *Avoid words always, never, completely, definitely.*
- *Avoid associations, identical or reflective sentences on stem.*
- *Avoid answer choice pairs that give clues to correct answer choices.*
- *Avoid awkward choices.*
- *Make sure the answer choices have the content and structure of the language equivalent.*

Create a distractor to make it possible to be selected.
Use common mistakes that test takers commonly
make as material for writing distractors.

Avoid the use of humor.

EXAMPLE OF WRITING FLUID REASONING ITEMS

Fluid reasoning (Gf):
The use of controlled and
deliberate procedures
(usually requiring
concentration) to
immediately solve new
problems that cannot be
solved using previously
learned habits.



Induction (penalaran induktif): the ability to observe a phenomenon and find the principles or rules underlying the phenomenon.



General sequential reasoning (penalaran deduktif): the ability to perform logical reasoning using premises and principles.



Quantitative reasoning (penalaran kuantitatif): the ability to perform reasoning using quantities, mathematical relationships, and mathematical operations.

Induction



(a)

(b)

(c)

(d)



(1)

(2)

(3)

(4)

Induction

SEPATU : KAKI

☐ Anting : Telinga

☐ Jari : Cincin

☐ Kepala : Topi

☐ Kuas : Cat

☐ Ruangan : Meja

General sequential reasoning

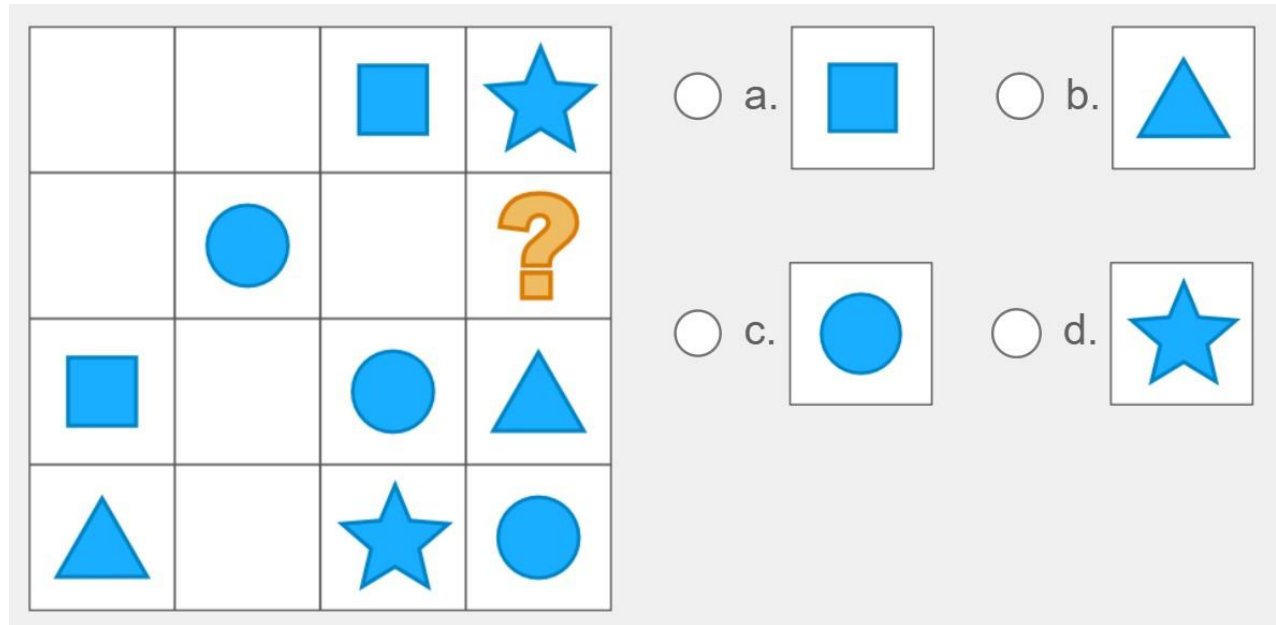
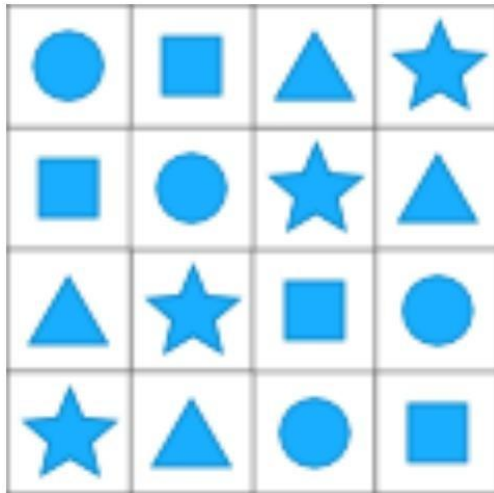
Semua bunga memiliki putik.

Semua Melati termasuk bunga.

- A. Semua melati memiliki putik.
- B. Semua melati tidak memiliki putik.
- C. Sebagian melati tidak memiliki putik.
- D. Sebagian bunga bukan melati.
- E. Tidak semua melati memiliki putik.

General sequential reasoning

- A row or column never contains the same shape twice.
- Each row and column contains the same type of shape.



Quantitative reasoning

2, 4, 6, 8, 10, ...

A. 9

B. 11

C. 12

D. 14

E. 15

Quantitative reasoning

Ali membeli sepatu yang sudah lama diinginkannya. Ia memperoleh harga lebih murah yaitu Rp 200.000,- karena ada diskon. Jika diskon yang diberikan adalah 20% dari harga normal, berapa harga sebenarnya dari sepatu tersebut?

- A. 160.000
- B. 220.000
- C. 225.000
- D. 240.000
- E. 250.000

ASSIGNMENT Meeting 7 – RTM 2



1. Explain the difference between a potential test (tes potensi) and achievement test (tes prestasi)
2. Compile a blueprint for achievement tests (materi bebas) and potential tests (materi bebas)
3. Each group wrote 5 points of achievement test and 5 points of potential test
4. Describe the capabilities measured by each item
5. Deadline: Meeting 8 (check project)